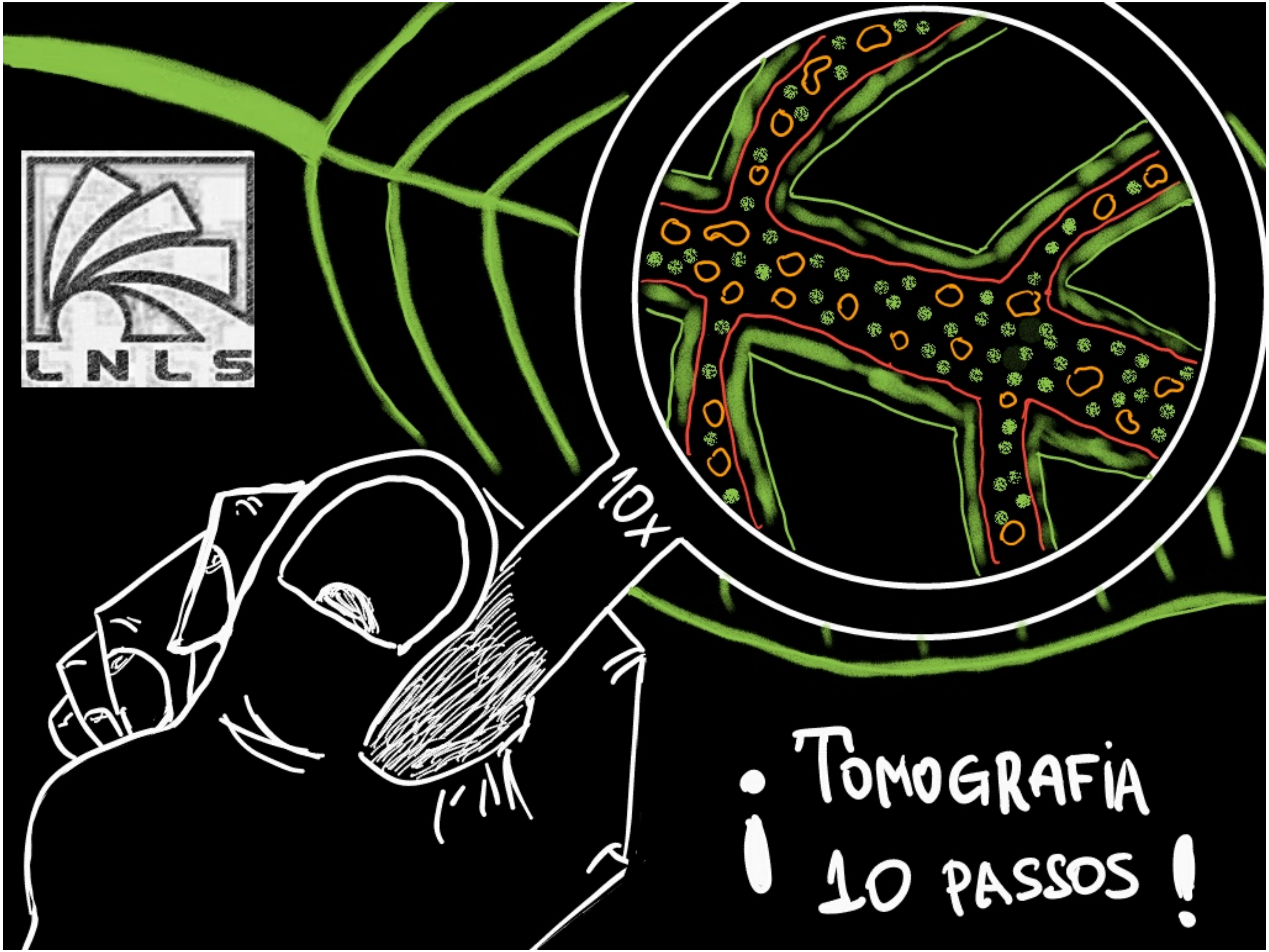
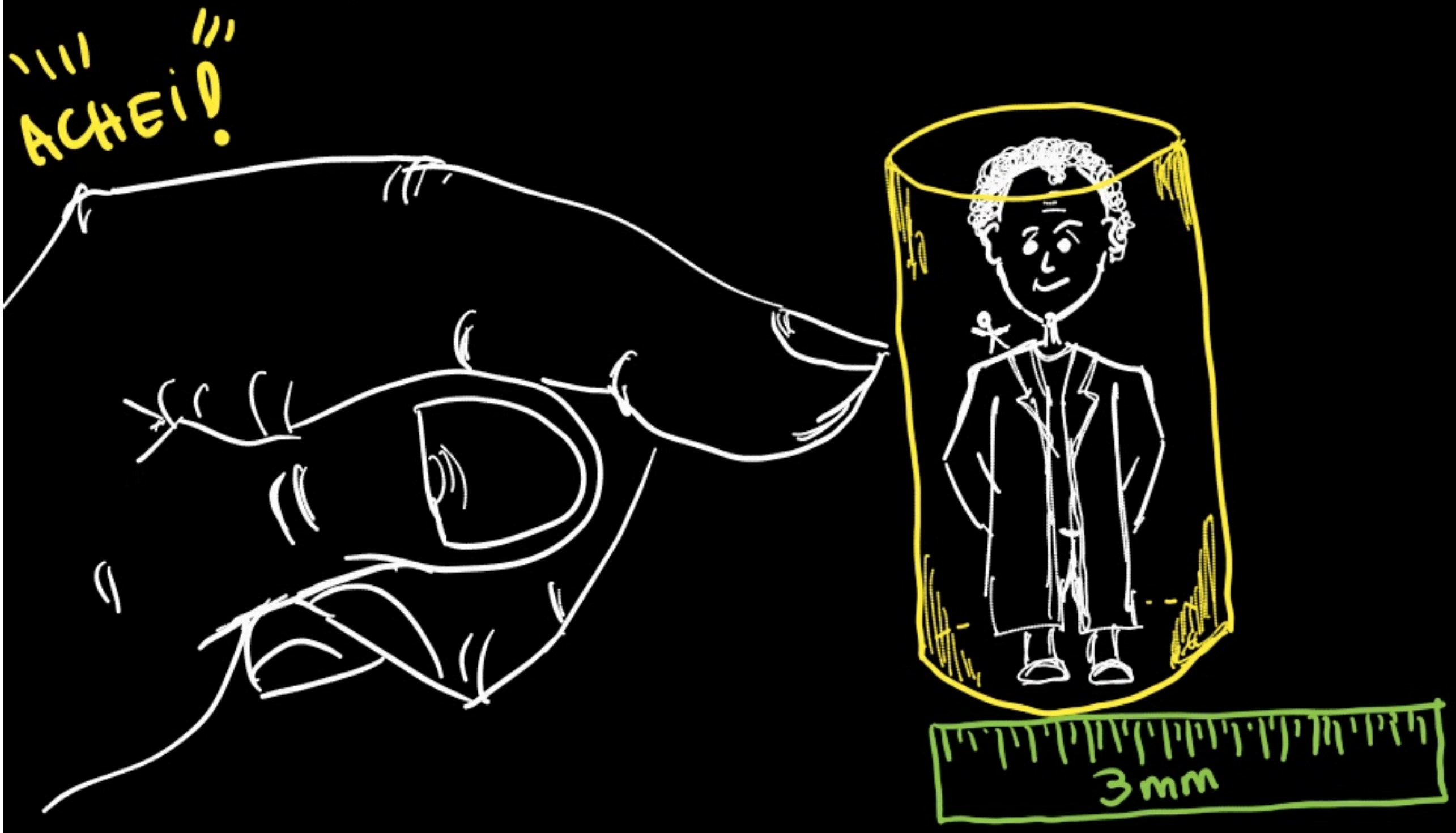

LNLS
10x
¡TOMOGRAFIA
10 PASSOS!

1. ESCOLHE UMA "AMOSTRA"

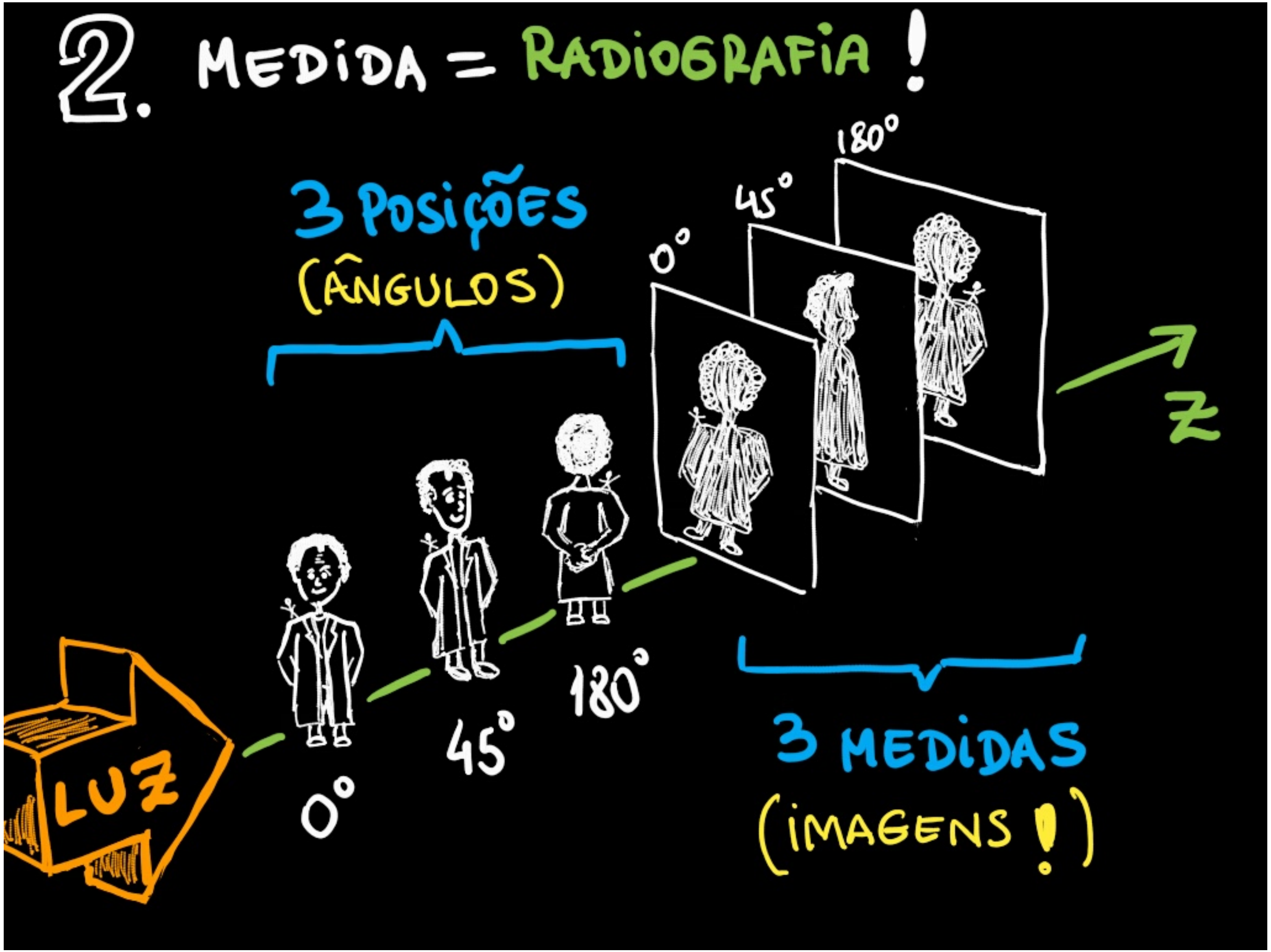
(CASO CIENTÍFICO!)



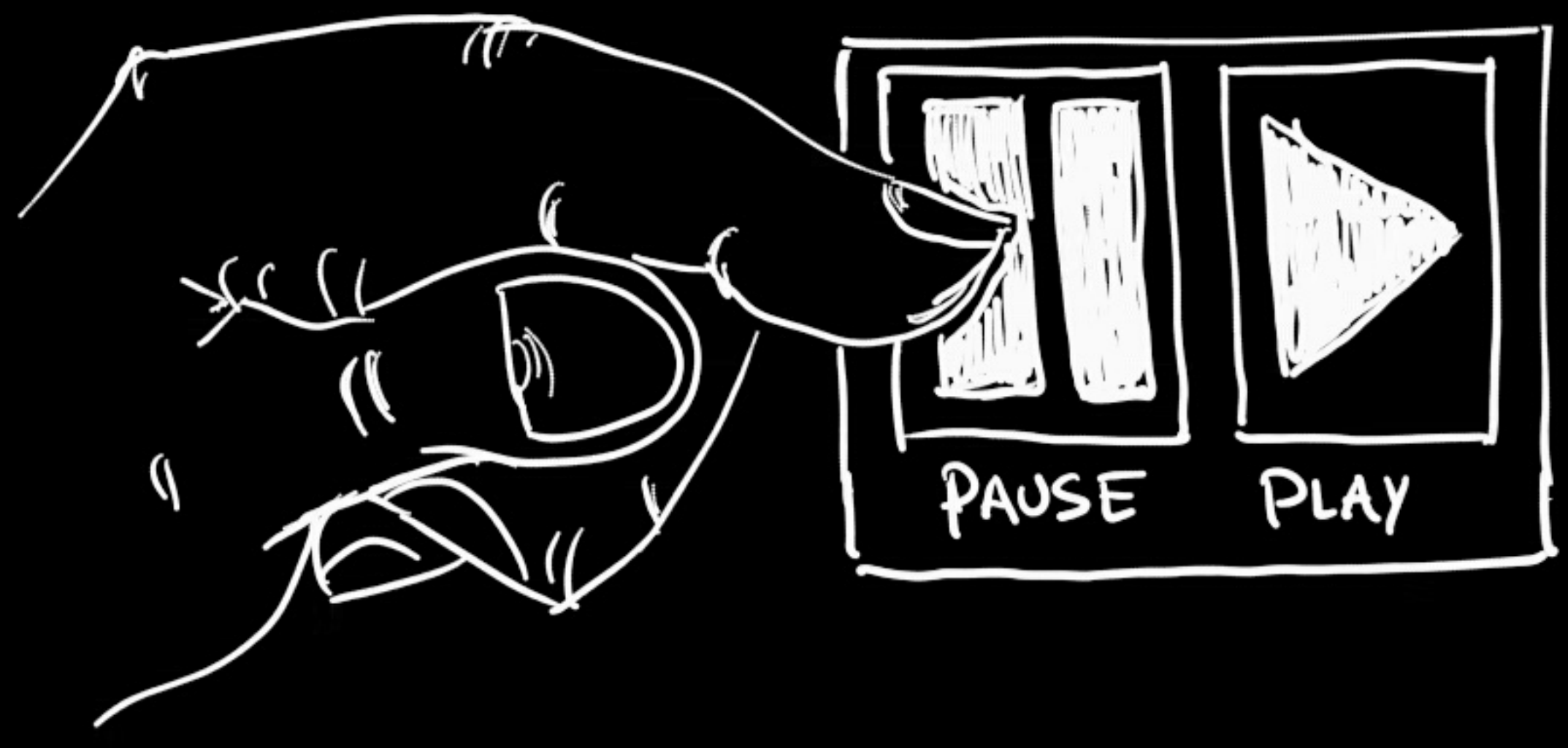
2. MEDIDAS POR ÂNGULOS !

0° ... 45° ... 180°

2. MEDIDA = RADIOGRAFIA !



... IMAGENS ?

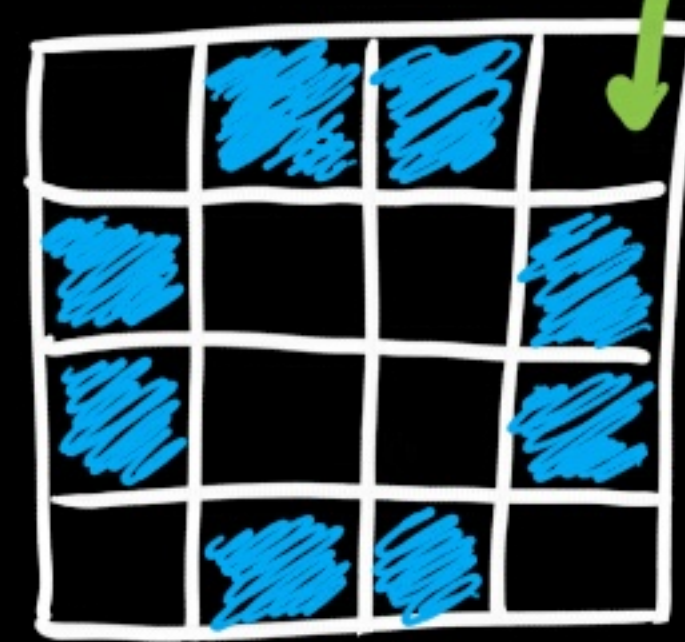


... IMAGEM TEM RESOLUÇÃO!

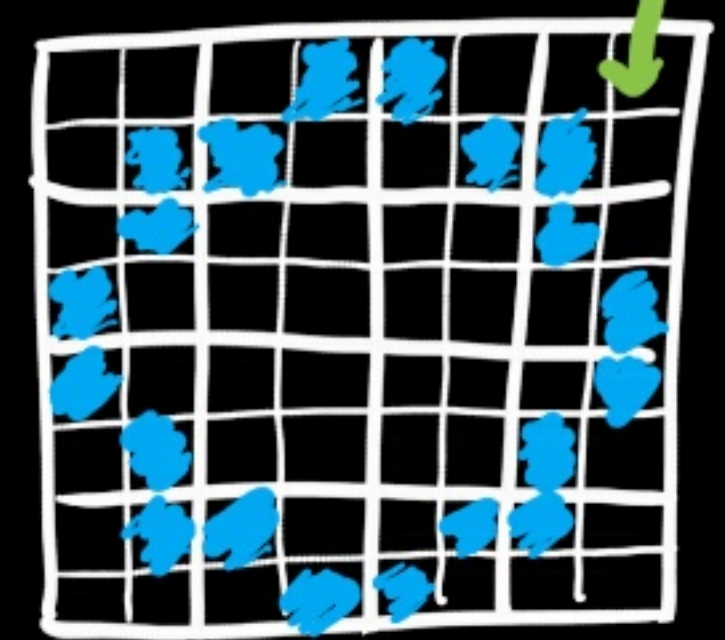
BEM RESOLVIDO

MAL RESOLVIDO

PIXEL



4x4



8x8

CÍRCULOS EM 2 RESOLUÇÕES!

... IMAGENS TEM INTENSIDADES!

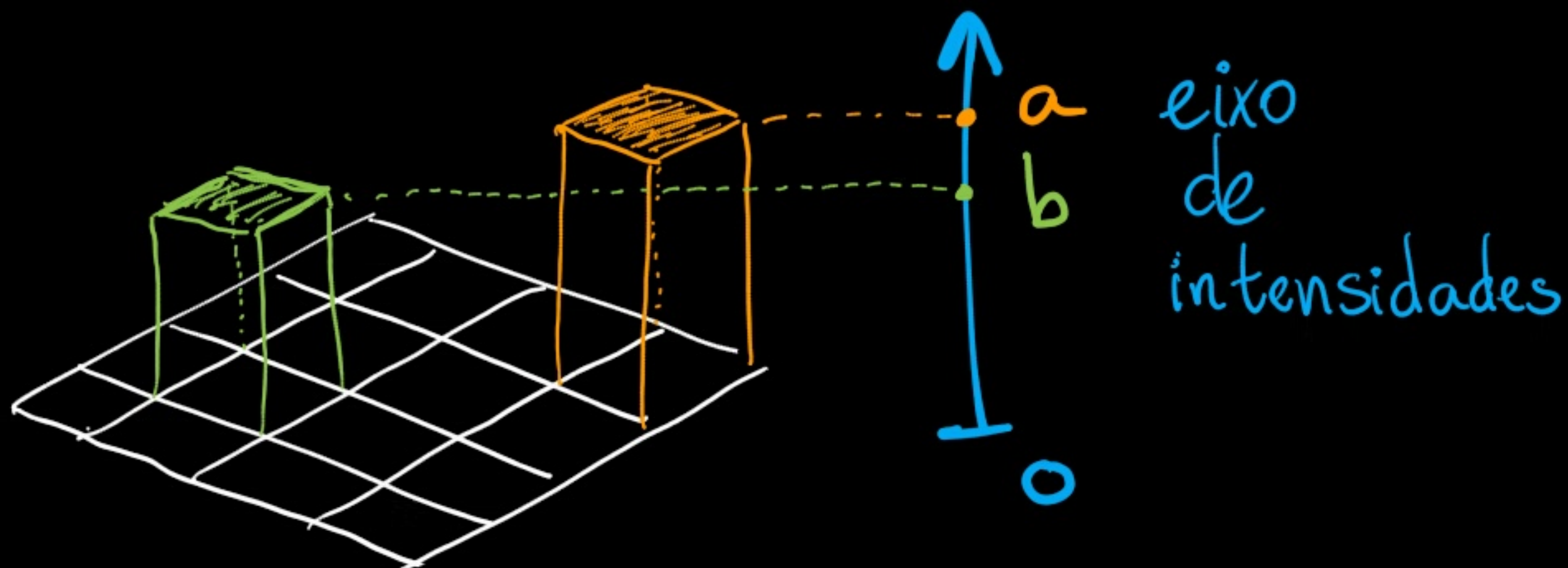
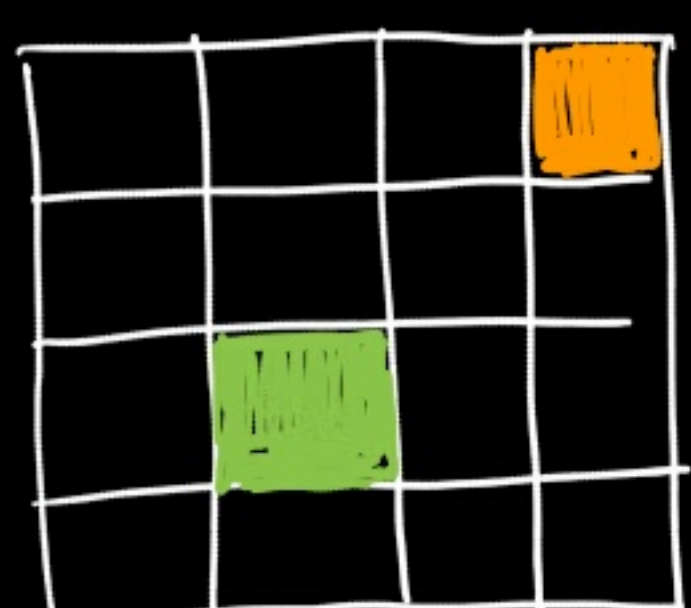
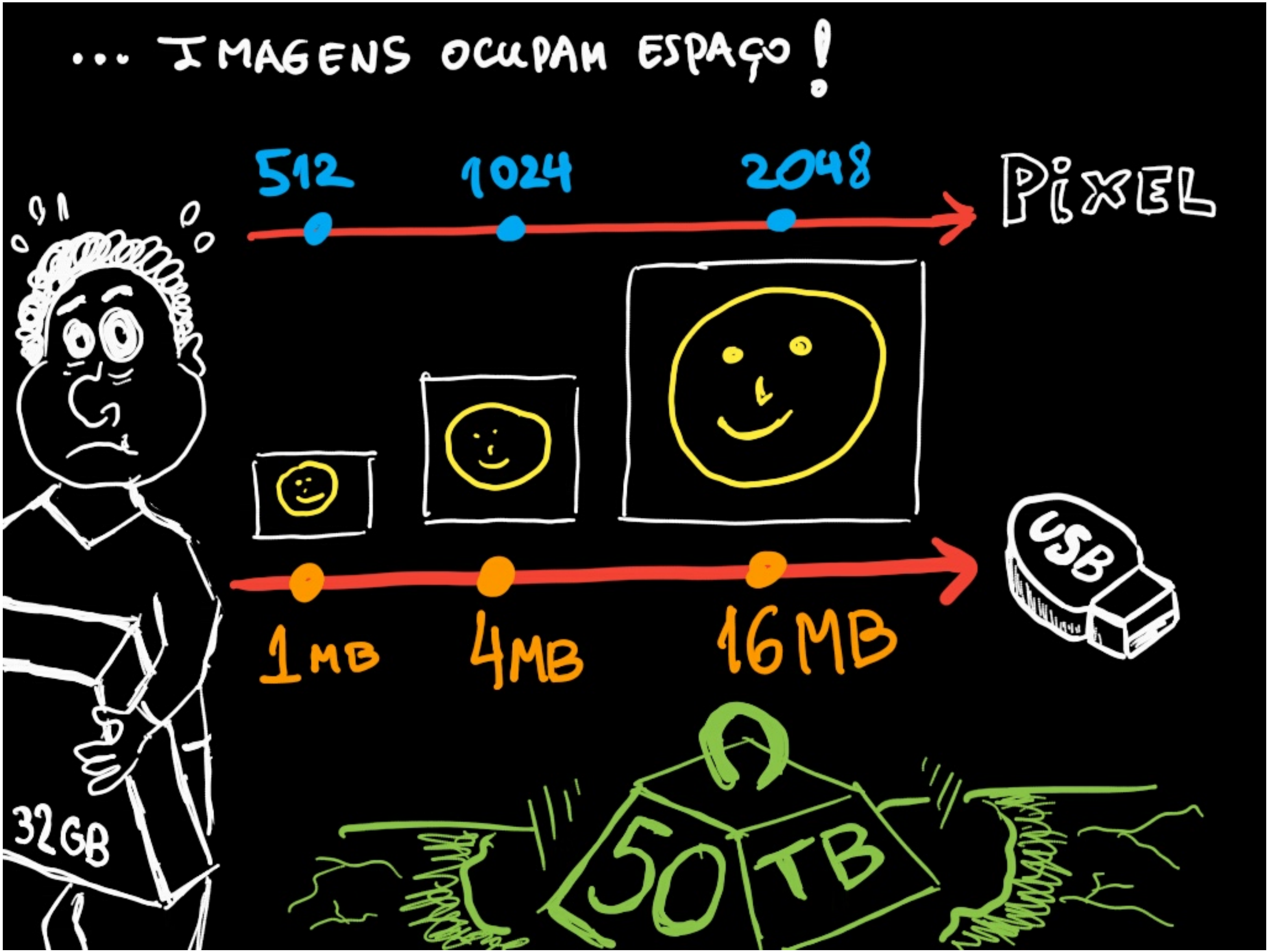


IMAGEM:

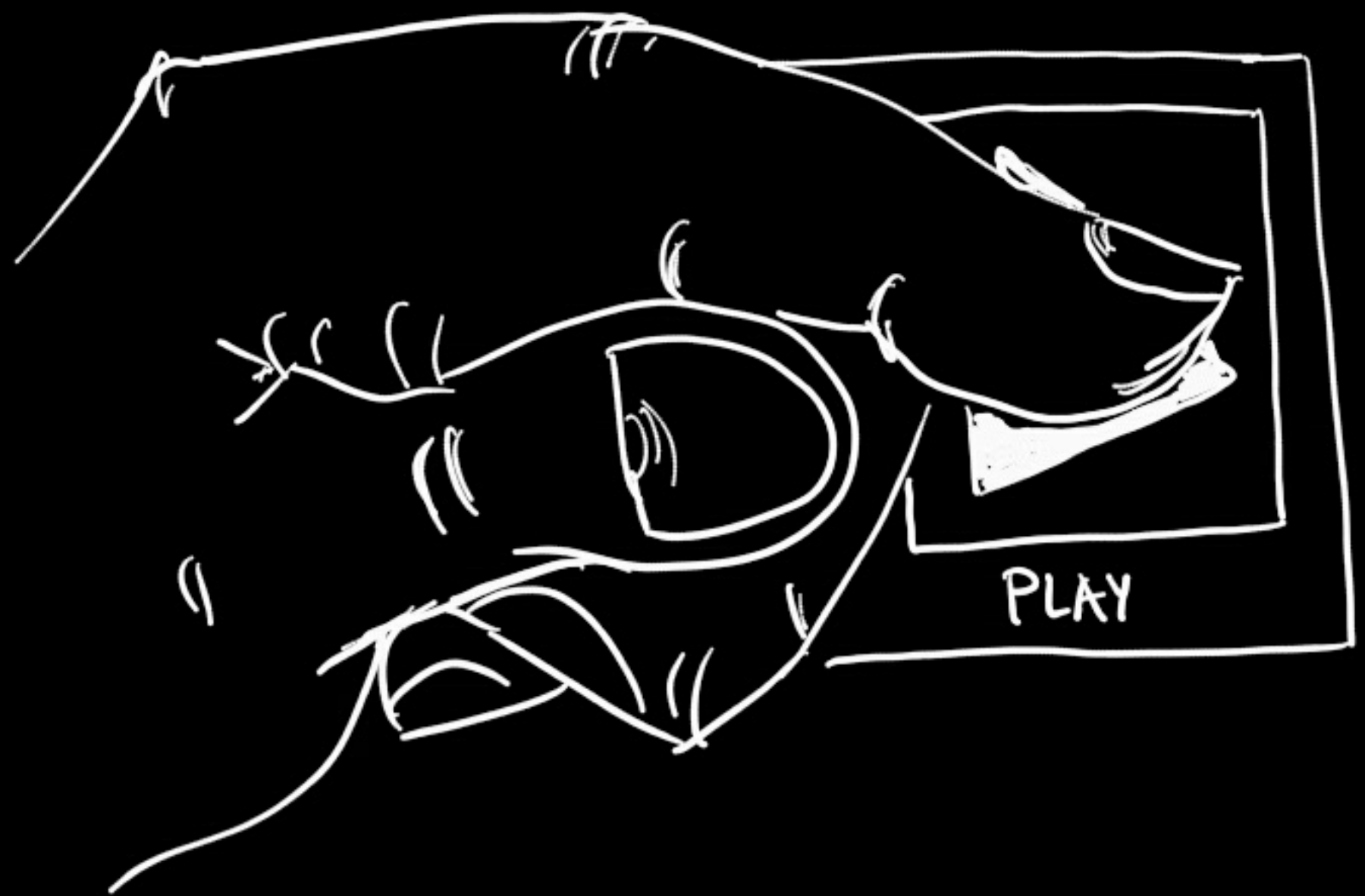


$a, b \in \mathbb{R}$

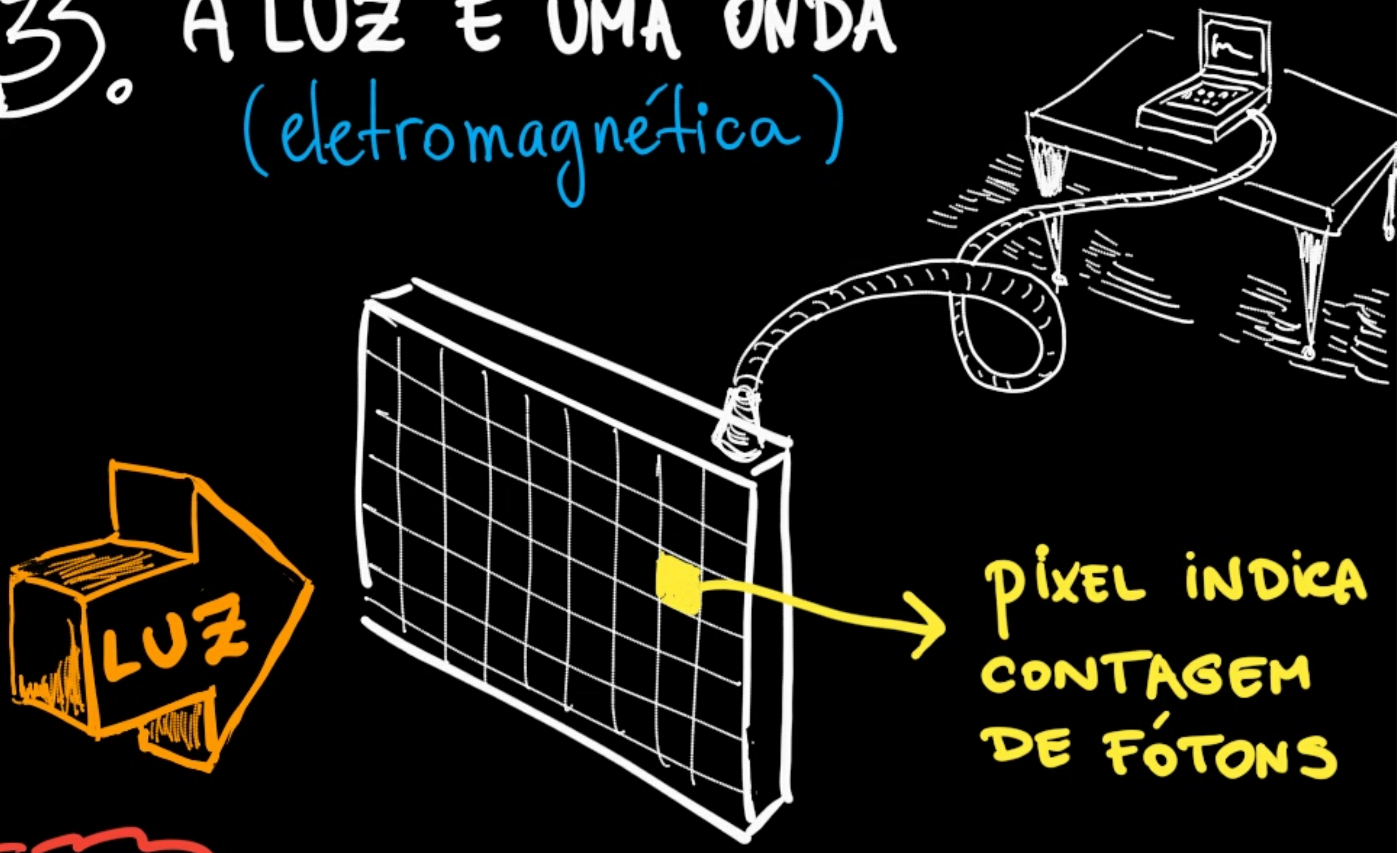
(cada pixel é um número)


... IMAGENS OCUPAM ESPAÇO!
512
1024
2048
PIXEL
1MB
4MB
16MB
USB
32GB
50TB

VOLTANDO...



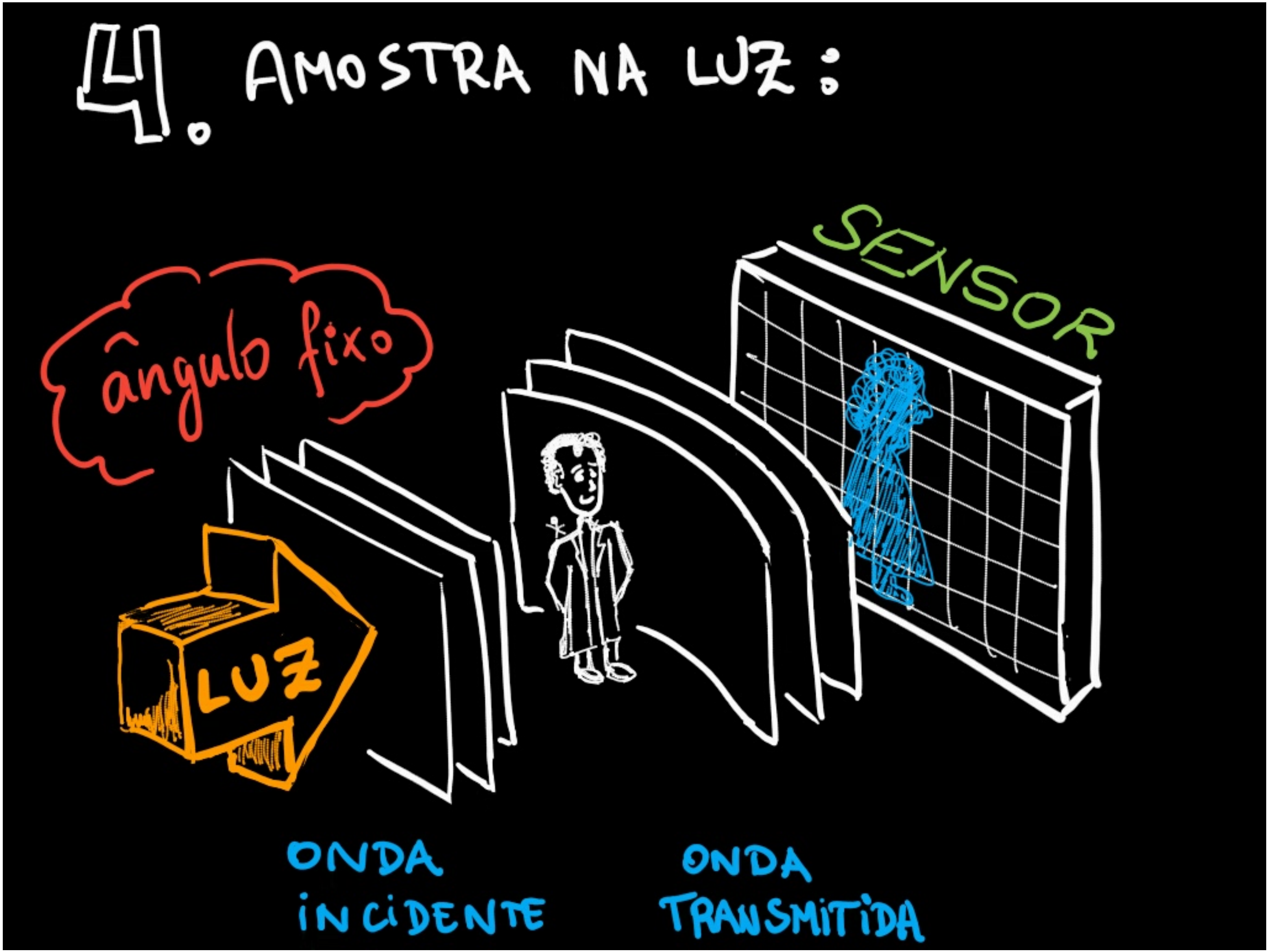
3. A LUZ É UMA ONDA (eletromagnética)



Podemos medir a luz

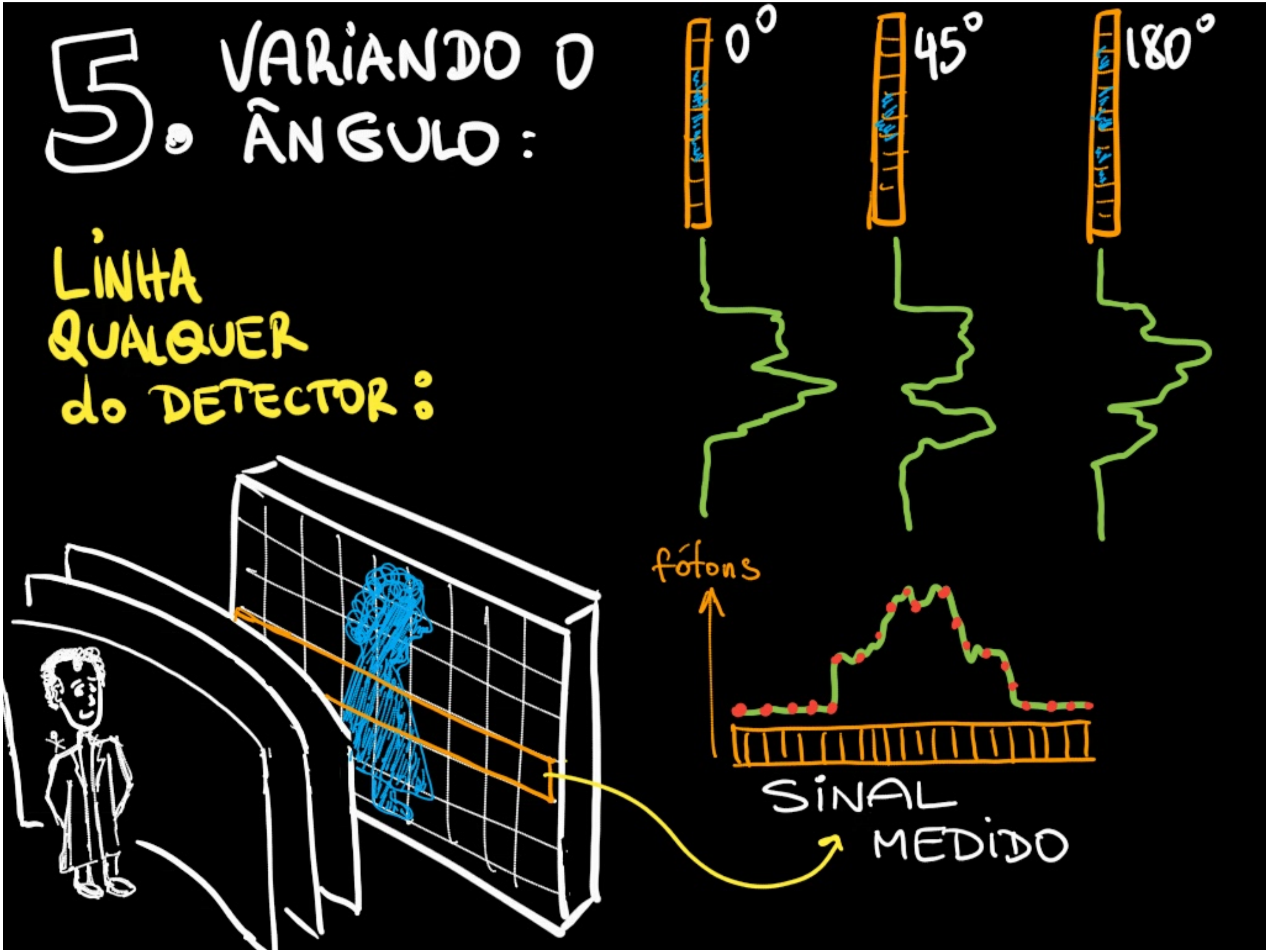
DETECTOR (UMA IMAGEM!)

4. AMOSTRA NA LUZ:

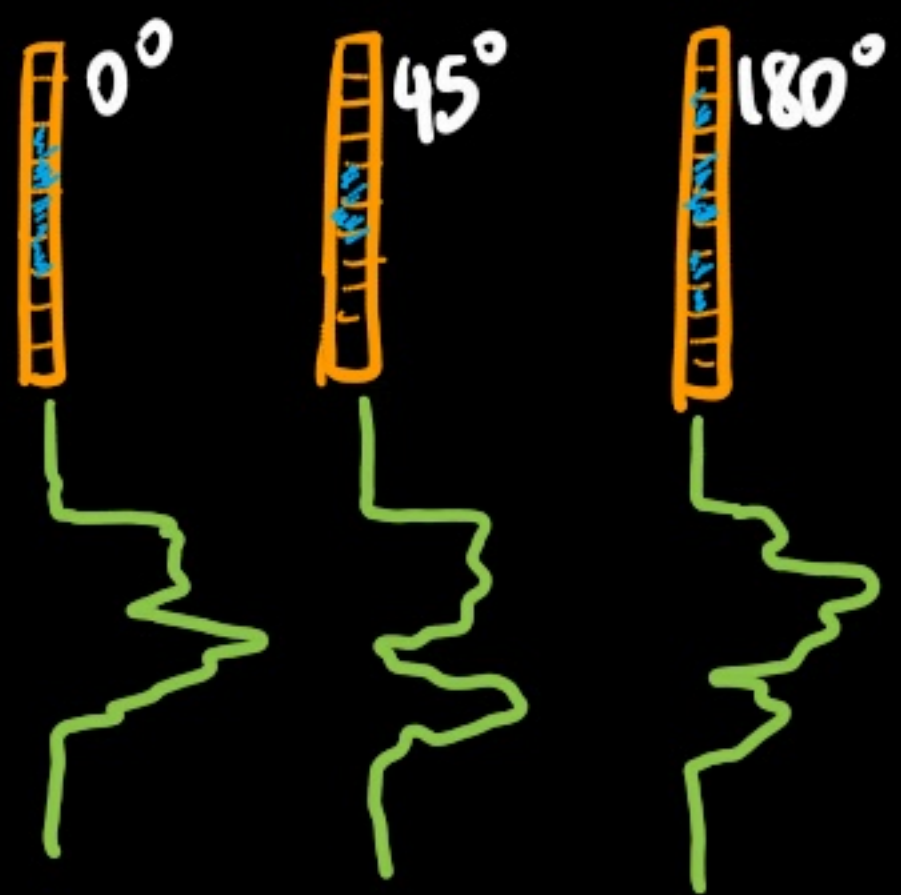


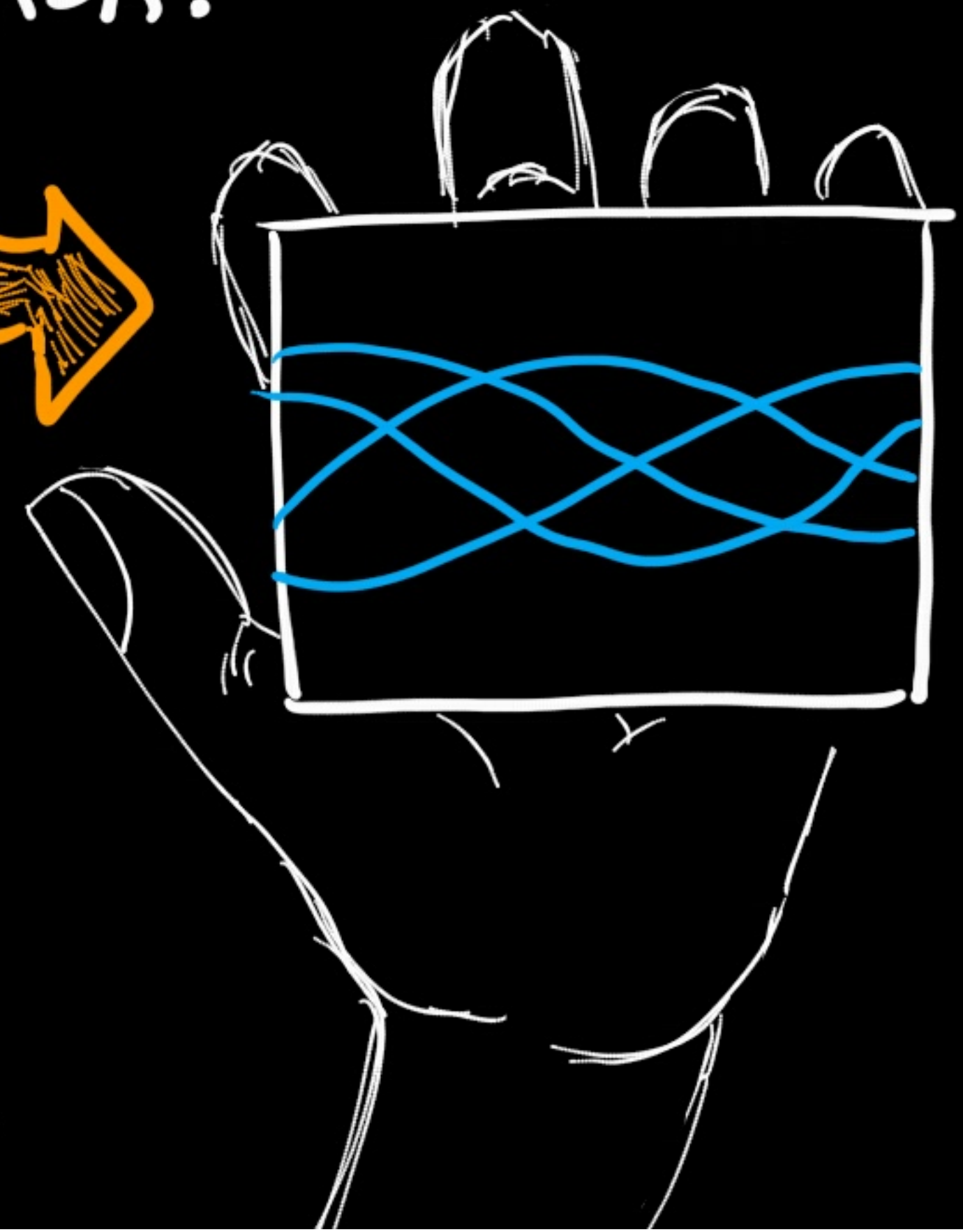
5. VARIANDO O ÂNGULO:

LINHA QUALQUER do DETECTOR:

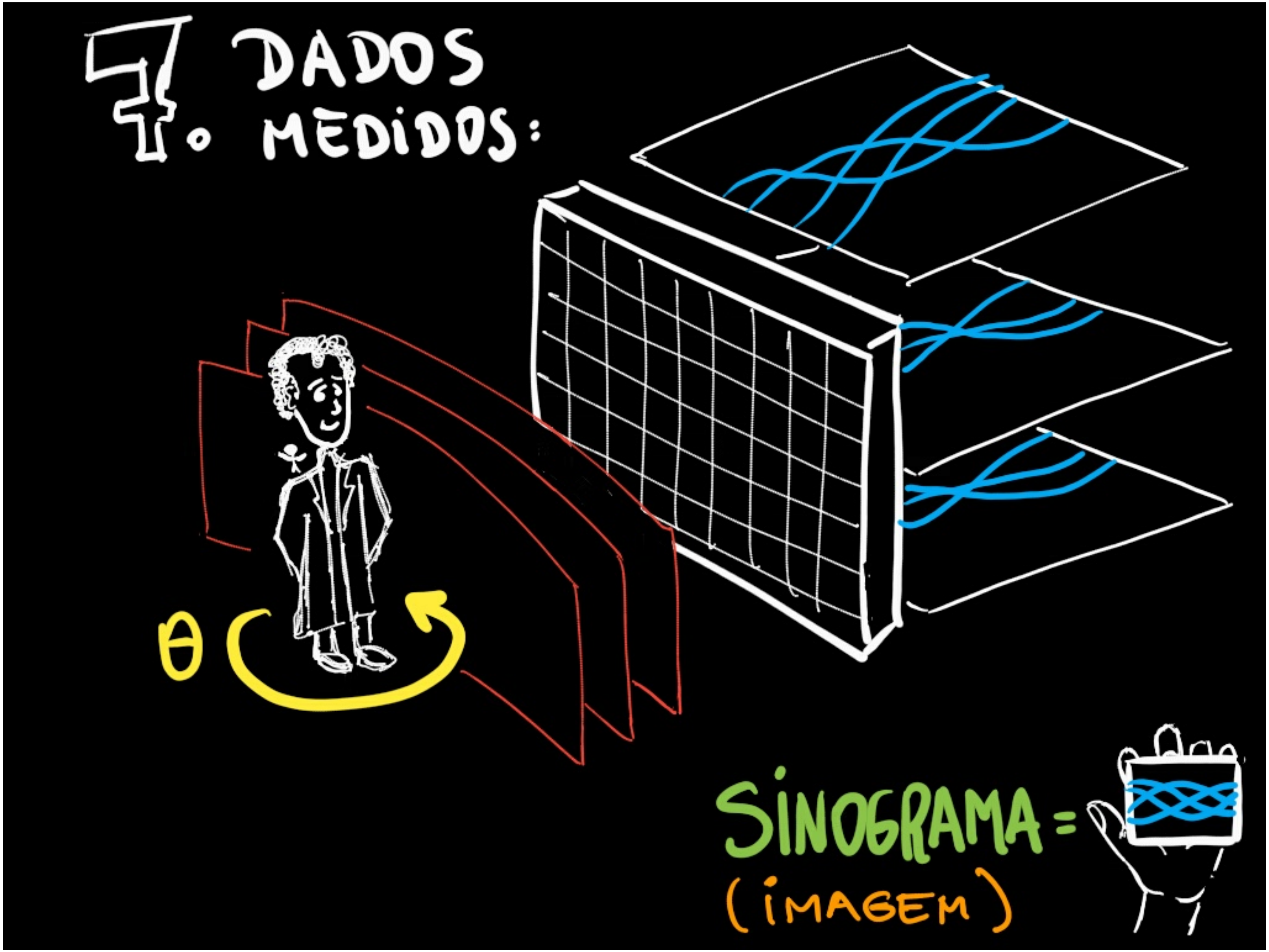


6. OUTRA IMAGEM É FORMADA:

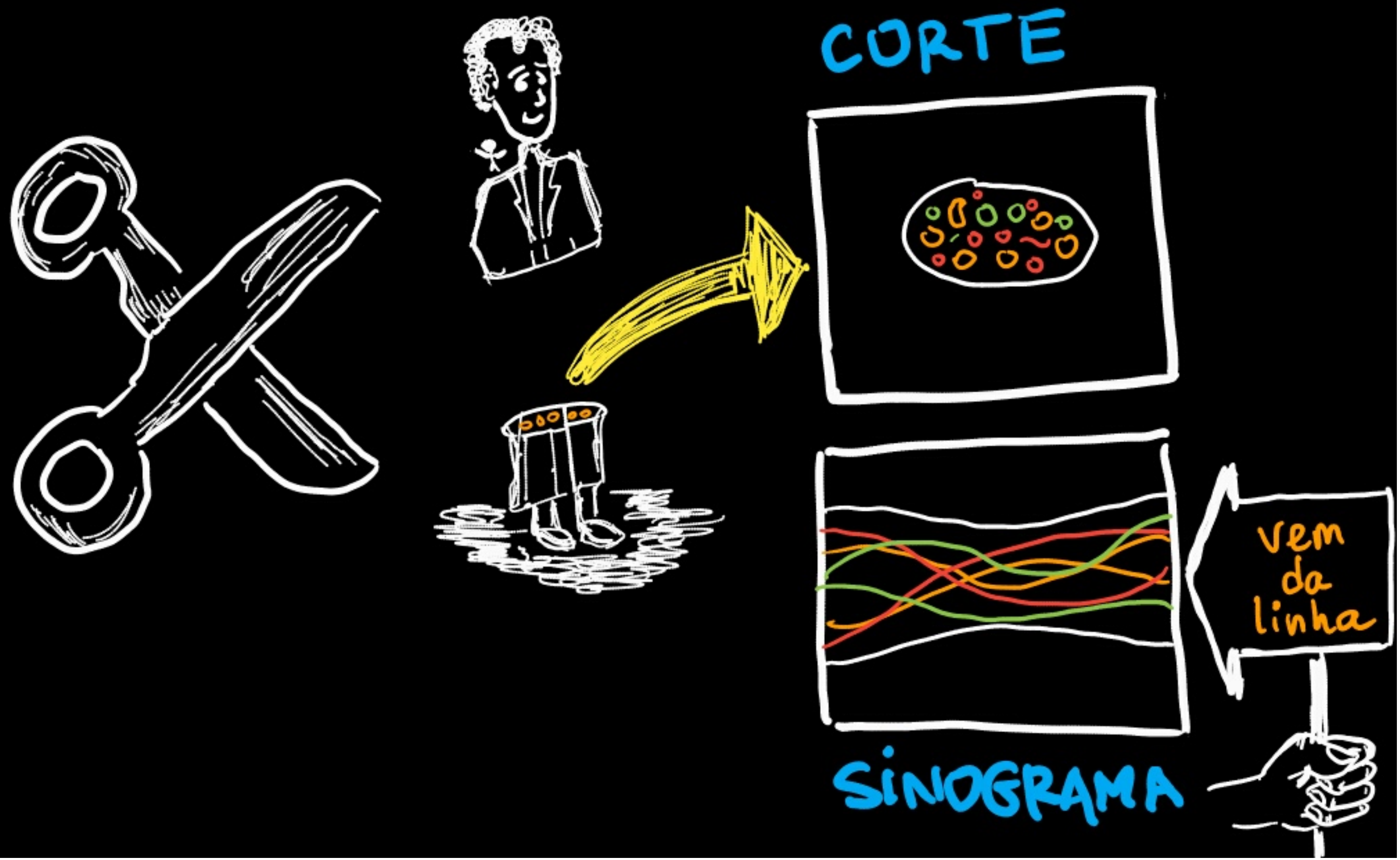




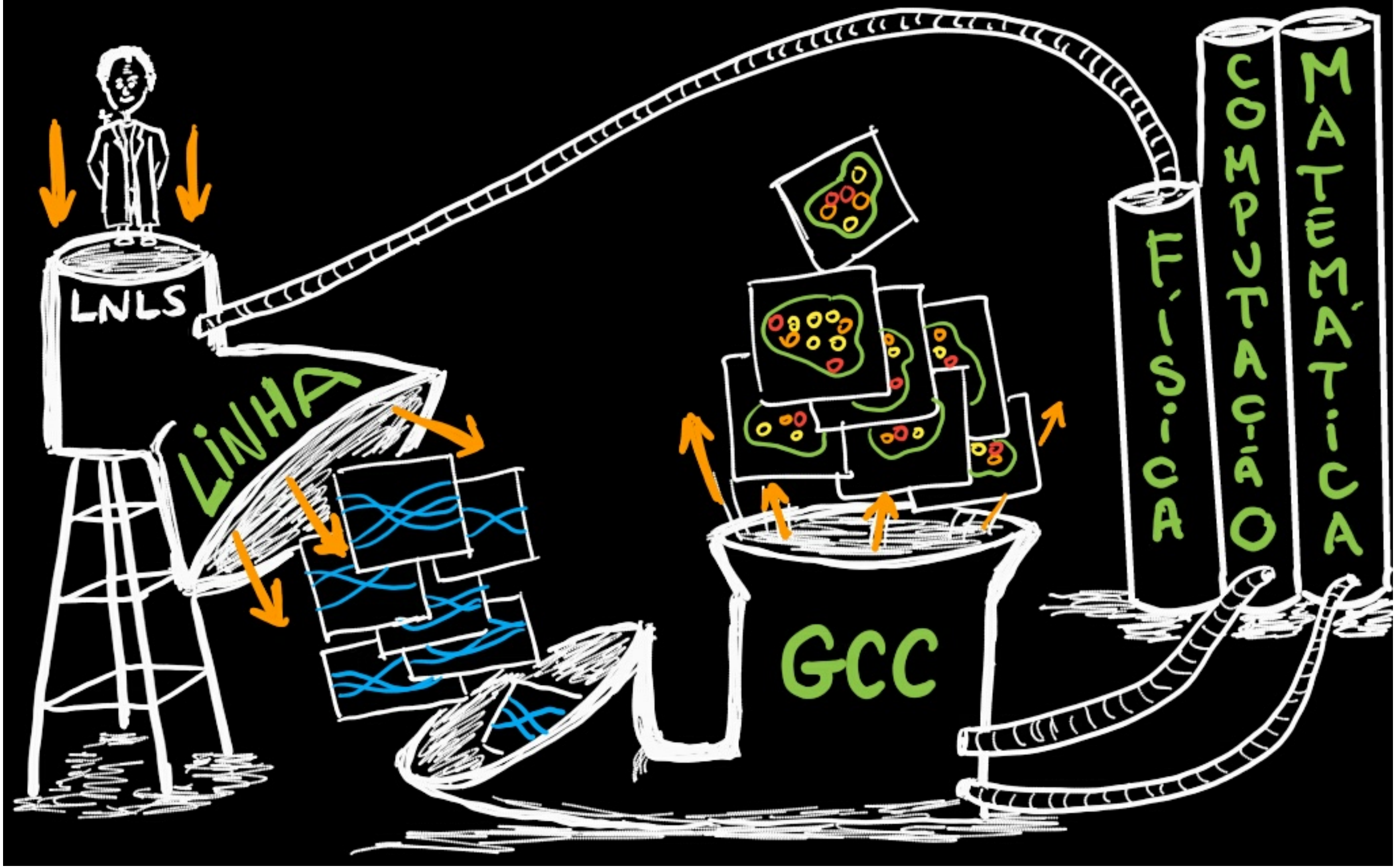
JUNTANDO TODOS OS SINAIS, PARA TODOS OS ÂNGULOS


DADOS
MEDIDOS:
θ
SINOGRAMA =
(IMAGEM)

8. CORTE TRANSVERSAL!



9. MÁQUINA DE IMAGENS!



10. ANÁLISE DAS IMAGENS !

OBRIGADO!
PODE PERGUNTAR.